The Choice of Data-Lock Anchor Changes Estimated Post-Completion Endpoint-Amendment Rates 4-fold on ClinicalTrials.gov: A Methodological Note With a Reusable Pipeline

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Abstract

Background. Registered-primary-outcome changes between trial data lock and results publication are a form of selective outcome reporting. Any prevalence estimate for such changes in `ClinicalTrials.gov` (CT.gov) depends on the temporal anchor chosen to define "data lock." To our knowledge, no prior audit has quantified how much that choice matters.
Objective. Estimate how the choice of data-lock anchor (`ResultsFirstPostDate` vs `ResultsFirstSubmitDate`) affects the rate of "post-completion" primary-outcome amendments in recently updated Phase 3 industry-sponsored trials on CT.gov.
Methods. We sampled the 1,000 most-recently-updated COMPLETED or TERMINATED Phase 3 industry-sponsored trials (enrolment ≥ 300) on CT.gov, queried 16 April 2026. For each trial we retrieved the complete version history through `api/int/studies/{NCT}/history`—a CT.gov endpoint that powers the public History tab but is not in the v2 API reference. This provides live per-trial access to the same historical data the AACT database aggregates [1]. Every meaningful primary-outcome amendment was classified into temporal windows relative to `primary_completion_date` and a chosen data-lock anchor.
Results. 275/1000 trials (27.5%, 95% CI 24.8–30.3%) had at least one meaningful primary-outcome amendment. Classified against `ResultsFirstPostDate`, 115/1000 (11.5%, 95% CI 9.7–13.6%) fell in the "between-window" (primary completion → results posting). Replacing the anchor with `ResultsFirstSubmitDate` reduced this to 28/1000 (2.8%, 95% CI 1.9–4.0%)—a 4.1-fold reduction attributable entirely to anchor choice. Independent case-study verification of three high-profile candidates (NCT04569084, NCT04738487, NCT03179631) showed that registry text-level audits are a lower bound: they miss manipulation at the statistical-analysis-plan (SAP) level.
Conclusions. Future systematic audits of registered-outcome changes should report both anchor variants. Registry text audits capture only text-level endpoint changes and cannot detect SAP-level manipulation; they should be viewed as complementary to FDA briefing-document audits, not a replacement. The audit pipeline is released under MIT licence.

Keywords: `ClinicalTrials.gov`; registered outcome; primary endpoint; outcome switching; trial integrity; statistical analysis plan; methodology.

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1 Introduction

Registration of clinical-trial primary outcomes before enrolment is a cornerstone of the ICMJE [2] and FDAAA [3, 4] trial-transparency regime. A registered outcome that is changed after data lock, but before the trial's results are made public, is a form of selective outcome reporting known to inflate effect estimates [5, 6]. The COMPare study [7] documented pervasive discrepancies between registered and published outcomes in 67 trials via manual comparison. More recent work [8, 9] has automated downstream analyses of results posting but has not focused on the mid-pipeline window in which amendments are most ambiguous: *after primary completion, before results submission.*

Any prevalence estimate for post-completion primary-outcome changes requires a temporal anchor for "data lock." Two natural candidates are exposed by CT.gov's metadata schema:

- `ResultsFirstPostDate`—the date CT.gov *posted* the results submission to the public interface;
- `ResultsFirstSubmitDate`—the date the sponsor *submitted* the results to CT.gov for review, typically several weeks earlier.

Sponsors prepare results for submission in a defined internal window following the primary completion date. Amendments made during this submission-preparation window are administrative rewording of the registered outcome to match the final results module, not manipulation of the endpoint definition itself. Classifying such changes as "post-completion manipulation" over-estimates the manipulation rate.

In this methodological note we ask: *how much does the choice of anchor matter?* We audit a 1,000-trial cohort from CT.gov and report rates under both anchors. We also illustrate, with three high-profile case studies, that text-level registry audits fundamentally cannot detect manipulation at the statistical-analysis-plan (SAP) level—a limitation that applies to all prior work in this area as well as our own. We release the pipeline as open source.

Contributions.

1. Empirical quantification of the anchor-choice sensitivity: 4.1-fold difference in the estimated "post-completion" amendment rate (from 11.5% to 2.8%) across 1,000 trials.
2. A reusable audit pipeline implemented against a publicly-accessible CT.gov endpoint (not documented in the v2 API reference but mirroring what AACT makes available in aggregated form [1]), enabling targeted per-trial audit without an AACT database clone.
3. Case-study illustration, via three high-profile trials, that text-level audits are a *lower bound* on manipulation—SAP-level changes (population switches, analysis-method changes) are invisible to registry text comparison. This is relevant not only to our pipeline but to any prior registry-text audit including COMPare.

2 Methods

2.1 Data source: CT.gov internal history endpoint

The CT.gov study page exposes a public "History of Changes" tab that lists every historical revision of the registration. This tab is populated by two internal endpoints:

- `GET /api/int/studies/{NCT}/history` returns a list of all versions with `version`, `date`, `status`, and `moduleLabels` indicating which protocol modules changed in that version.
- `GET /api/int/studies/{NCT}/history/{version}` returns the full protocol and results section for that specific historical version.

These endpoints are stable and publicly accessible but are not listed in the CT.gov v2 API reference. The same historical data is available in aggregated form through the AACT database [1]. To verify that the internal endpoint exposes consistent data with the public v2 API, we implemented a unit test (`tests/test_api_agreement.py`) that fetches the latest version from the internal endpoint and compares the primary-outcome list to the v2 API's current record for the same trial. All five tested trials (NCT04006457, NCT04991935, NCT03179631, NCT04738487, NCT02477800) showed identical primary-outcome sequences. Our pipeline exists to enable targeted per-trial audit without an AACT database clone; it is not intended to replace AACT for systematic work.

2.2 Cohort

We queried CT.gov on 16 April 2026 with filters `overallStatus=COMPLETED,TERMINATED`, `Phase=PHASE3`, `LeadSponsorClass=INDUSTRY`, `enrollmentCount` ≥ 300, sorted by `LastUpdatePostDate` descending, and retained the first 1,000 records.

2.3 Primary-outcome change classifier

For every trial we retrieved the subset of versions whose `moduleLabels` indicated a change to the Outcome Measures module, plus `version 0` (initial registration) and the latest version. For each adjacent pair we computed the symmetric set difference of normalised primary-outcome texts.

Normalisation lowercased the text, retained parenthetical abbreviations (e.g., "(Cmax)"), stripped common analytic prefixes ("change from baseline in ..."), and expanded a small dictionary of field-specific acronyms (CDR-SB, ALSFRS-R, PFS, OS, ORR, DoR, PANSS, MADRS). Two measures were considered the same if either (a) their content-token Jaccard similarity exceeded 0.5, or (b) one was a superstring of the other at $\geq$ 75% content-token containment. Thresholds were fixed *a priori* and were not calibrated against a labelled ground-truth endpoint-swap dataset; sensitivity analysis is deferred (see Limitations).

Any outcome added or removed after normalisation was a *meaningful* change; changes that were detected only as count shifts (without text shift) were *not* counted.

2.4 Temporal windows

Let c be the date of the amendment, p the trial's `primaryCompletionDate`, s its `ResultsFirstSubmitDate`, and r its `ResultsFirstPostDate`. Two anchor conventions were evaluated:

v1 anchor (`ResultsFirstPostDate`, naïve).

$$\begin{aligned} \text{A}: &\ c < p, \\ \text{B}: &\ p \le c < r - 30\text{d}, \\ \text{C}: &\ r - 30\text{d} \le c \le r + 30\text{d}. \end{aligned}$$

v2 anchor (`ResultsFirstSubmitDate`).

$$\begin{aligned} \text{A}: &\ c < p, \\ \text{B}: &\ p \le c < s - 7\text{d}, \\ \text{C}: &\ s - 7\text{d} \le c \le r + 30\text{d}. \end{aligned}$$

The 7-day buffer before submission is a pragmatic convention to avoid bordering single-day coincidences; it is not derived from ICMJE or FDAAA policy. When `ResultsFirstSubmitDate` was unavailable (results not yet submitted) we conservatively treated such trials as uncensored B candidates.

2.5 Statistical reporting

Proportions are reported with Wilson 95% confidence intervals [10]. Differences between v1 and v2 B-window rates are reported as raw counts and ratios; no formal hypothesis test is performed because the comparison is between two classifiers applied to the same cohort (not two samples).

2.6 Case-study verification protocol

Three high-profile candidates from the v1 B-window list (NCT04569084 Tanabe oral edaravone; NCT04738487 Merck pembrolizumab/vibostolimab; NCT03179631 PTC ataluren) were investigated by web search (press releases, peer-reviewed publications, regulatory documents, independent trade commentary) to determine whether the registered change corresponded to a substantive manipulation or to an administrative amendment. Case-study verification was performed by a single investigator; no inter-rater assessment is reported. The three cases are *illustrative*, not a random sample.

3 Results

3.1 Overall amendment rates

Table 1 summarises the 1,000-trial audit. Of all trials, 275 (27.5%, 95% CI 24.8–30.3%) had at least one meaningful primary-outcome amendment across their version history.

3.2 The anchor-choice effect

The key result is the 4.1-fold difference between the v1 and v2 B-window rates (11.5% vs 2.8%, ratio = 4.11; Figure 1). Of the 115 v1 B-hits, 87 (75.7%, 95% CI 66.9–82.6%) reclassify as C under v2, and 28 (24.3%) remain in B.

The distribution of amendment dates relative to the submission timestamp (Figure 2) shows the expected pattern: a dense cluster of text changes in the 0–60 day window immediately

Table 1. Primary-outcome amendment counts in 1,000 Phase 3 industry-sponsored CT.gov trials. Proportions with Wilson 95% CI.

Window	n	% of 1,000	95% CI
Any meaningful change	275	27.5	(24.8, 30.3)
A — pre-completion amendment	132	13.2	(11.2, 15.4)
B v1 — `ResultsFirstPostDate` anchor	115	11.5	(9.7, 13.6)
B v2 — `ResultsFirstSubmitDate` anchor	**28**	**2.8**	(1.9, 4.0)
C — results-reporting rewording	83	8.3	(6.7, 10.2)

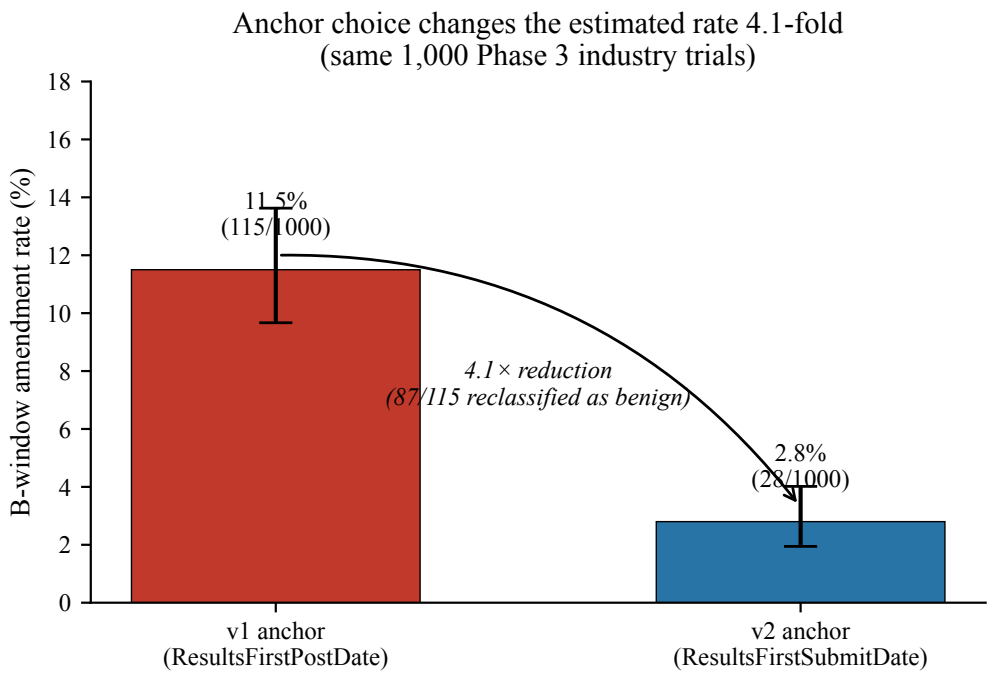

Figure 1. Estimated B-window amendment rate under the two anchor conventions. Error bars are Wilson 95% confidence intervals. The ratio is deterministic once the cohort is fixed; the difference captures amendments made during the sponsor's results-submission preparation window.

preceding submission, tapering toward the tail. The v2 classifier assigns the cluster to C (benign rewording during results preparation) and retains only the sparse long-tail amendments in the B-window.

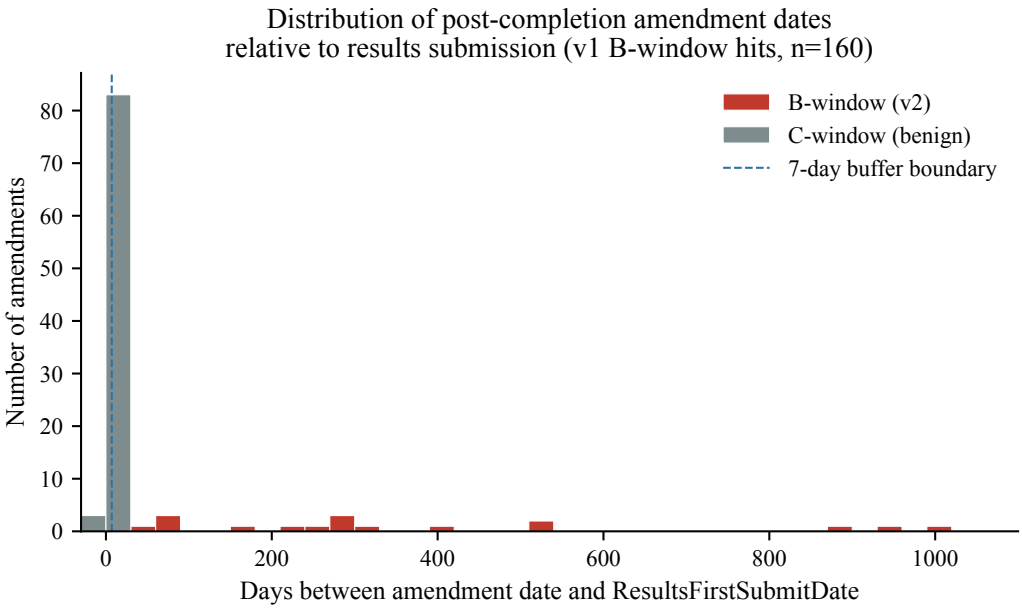

Figure 2. Distribution of meaningful primary-outcome amendment dates relative to `ResultsFirstSubmitDate`, across the 160 individual amendment events contributed by the 115 v1 B-window trials (some trials contributed more than one amendment event). The submission-preparation cluster (right-edge red/grey stack) drives the 4-fold inflation in the v1 rate.

This ratio is *deterministic* given the anchor choice; it is not a random-sample finding. The interpretation is that sponsors amend registered primary-outcome text during the results-submission preparation window at a rate that dwarfs genuine post-completion manipulation, and any audit that classifies such amendments as "post-completion" without using a submission-date anchor will over-estimate the manipulation rate by approximately 4-fold.

3.3 Three case-study verifications

Three v1 B-window trials with high public profiles were investigated to illustrate the interpretive gap between a flagged registry change and substantive trial manipulation.

NCT04569084 — Tanabe oral edaravone (ALS). Our classifier flagged a primary-outcome change from *ALSFRS-R at Week 48* to *CAFS at Week 48* in version 56 (2024-11-05). Investigation established that the trial was terminated for futility on 31 July 2023 on the ALSFRS-R endpoint [11]; final results were presented 17 June 2024 and published September 2025 [12] reporting both CAFS (primary) and ALSFRS-R (key secondary), both negative. The registry amendment post-dated the publication by four months and represents retroactive alignment of the registration module, not hidden manipulation. **Verdict: registry lag; no manipulation.**

NCT04738487 — Merck pembrolizumab/vibostolimab (KEYVIBE-003). Our classifier flagged a primary-outcome collapse from five listed primaries (PFS and OS across three PD-L1 TPS bins) in version 45 to one primary (OS in TPS $\geq$ 50%) in version 136, post primary completion. Investigation established [13] that the protocol's pre-specified primary was always

OS in TPS $\geq 50\%$; version 45's five-primary listing was an earlier registration error. Merck announced on 16 December 2024 that the trial stopped for futility and that the entire vibostolimab programme (KEYVIBE-003, -006, -007, -008, -010) was discontinued. **Verdict: registry error correction; no manipulation.**

NCT03179631 — PTC ataluren (Duchenne muscular dystrophy, Study 041). Our classifier flagged an expansion from one primary outcome (*slope of 6MWD change over 72 weeks*) in version 10 to four primaries (ITT vs mITT $\times$ change vs average rate) in version 50, 3.8 years after primary completion. Investigation established that the published paper [14] reports one primary (slope in ITT, $p = 0.0248$), and the four-endpoint listing reflects CT.gov results-module formatting rather than a scientific re-analysis. However, a substantive manipulation did occur at a layer our pipeline cannot see: the statistical analysis plan's pre-specified analysis *population* was switched between earlier submissions and the final analysis. The FDA's reviewing division withdrew the NDA resubmission on 13 February 2026 stating that the data were "unlikely to show enough efficacy to support regulatory approval" [15]; the EMA finalised non-renewal of the conditional marketing authorisation on 28 March 2025. **Verdict: registry text change cosmetic; real manipulation occurred at the SAP level and was invisible to text-level registry audit.**

4 Discussion

4.1 What this study establishes

The choice of data-lock anchor produces a 4.1-fold spread in the estimated rate of "post-completion" primary-outcome amendments. Using `ResultsFirstPostDate` conflates genuine post-completion changes with administrative rewording in the results-submission preparation window; using `ResultsFirstSubmitDate` separates the two. The 2.8% v2 rate is our best per-trial estimate of amendment activity in the submission-preparation and post-submission window, limited to the exact sampled population (recently-updated, recent-era, large Phase 3 industry trials).

4.2 What this study does not establish

The 2.8% figure is *not* a manipulation rate. A single-rater review of the three highest-profile v2 candidates found two registry- lag cases and one that masked substantive manipulation at a layer invisible to registry text comparison. A defensible manipulation rate requires (a) a random sample of flagged cases, (b) blinded adjudication by at least two independent raters, (c) pre-registered triage criteria, and (d) cross-reference to FDA briefing documents. These are appropriate targets for a follow-up study.

4.3 Generalisability

Our cohort was sorted by `LastUpdatePostDate` descending. This biases the sample toward recent, actively-maintained registrations (2023–2026 era). Generalisability to older trials, to trials that stopped updating after results posting, or to non-industry sponsors is unknown. A random-sample replication stratified by completion year is a natural extension.

4.4 Relation to prior work

COMPARE [7] compared 67 trials' registrations to their publications by manual inspection and found frequent outcome switching. Chen et al. [16] analysed results publication across academic medical centers. Zarin et al. [9] reported on a decade of results submission to CT.gov. Our study differs in scope and unit of analysis: we compare registration versions to each other across 1,000 trials and do not attempt registration-vs-publication comparison. The methods are complementary. AACT [1] exposes the same historical registration data we use, but as a relational database clone; our pipeline exists to enable targeted per-trial queries without local database setup, not to replace AACT's systematic infrastructure.

4.5 Limitations

1. Sampling by `LastUpdatePostDate` over-represents conscientious sponsors.
2. Case-study verification was performed by a single investigator without inter-rater reliability measurement.
3. Similarity thresholds (Jaccard 0.5, superstring 0.75) were not calibrated against a ground-truth endpoint-swap dataset; sensitivity analysis is deferred.
4. `primaryCompletionDate` values used for the A/B classification were the current-API values, not the historical values at each amendment time. Retrospective amendments to the completion date itself could shift the classification (Goldacre et al. document this practice [7]).
5. The 7-day buffer before `ResultsFirstSubmitDate` is pragmatic, not policy-grounded.
6. Text-level registry audits cannot detect statistical-analysis-plan (SAP) changes such as population switches or analysis-method changes. Complete manipulation detection requires access to SAP documents or FDA briefing materials.

4.6 Recommendations for future audits

- Report both anchor variants, not just one.
- Use `ResultsFirstSubmitDate`, where available, as the primary anchor; fall back to `ResultsFirstPostDate` only when submission date is unreported.
- Combine registry-text auditing with SAP-level auditing (FDA briefing documents where available).
- Pre-register triage criteria, sample size, and inter-rater assessment protocol before extending beyond the proof-of- concept stage.

Data and code availability

All audit code and classifier implementation are available at `https://github.com/shizhigu/ct-endpoint-audit` under the MIT licence. The full 1,000-trial audit output (JSON with per-version primary-outcome diffs and per-trial metadata) is archived on Zenodo (DOI: 10.5281/zenodo.19615594). The CT.gov internal history endpoint is accessed without authentication at `https://clinicaltrials.gov/api/int/studies/{NCT}/history`.

Conflicts of interest

The author declares no competing interests. This research was unfunded.

Acknowledgements

Automated critique tooling was used during manuscript preparation to identify methodological weaknesses in an earlier draft, including the tautological headline claim, the over-claimed "undocumented API" contribution, and the selection-bias issue in the sampling frame. The final text reflects those revisions.

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Supplementary Materials

Table 2. Supplementary Table S1. The 28 trials flagged as B-window (locked-data, pre-submission) under the refined anchor. Sorted by primary completion date (most recent first). *(none)* in the *Results-first submitted* column indicates results not yet submitted at query time; such trials remain B-window candidates pending future data.

NCT	Sponsor	Primary completion	Results-first submitted	Change date
NCT06284915	Sanofi Pasteur, a Sanofi Company	2025-11-14	(none)	2025-11-24
NCT05842954	Novartis Pharmaceuticals	2025-06-27	(none)	2025-12-11
NCT06553456	Protgen Ltd	2025-05-14	(none)	2025-09-24
NCT05811247	Apnimed	2025-05-07	(none)	2025-11-18
NCT05997290	BioNTech SE	2025-03-13	(none)	2026-03-09
NCT05813275	Apnimed	2025-03-10	(none)	2025-11-18
NCT05743075	Nuance Pharma (shanghai) Co., Ltd	2024-12-06	(none)	2026-01-26
NCT04738487	Merck Sharp & Dohme LLC	2024-09-05	2025-07-28	2024-10-28
NCT04777201	Hoffmann-La Roche	2024-09-03	2025-09-02	2024-12-06
NCT04770220	Alzheon Inc.	2024-05-29	2025-09-11	2024-07-31
NCT05648110	AstraZeneca	2024-03-29	(none)	2025-09-25
NCT05573464	AstraZeneca	2024-03-26	2025-09-02	2025-03-25
NCT05142774	Organon and Co	2024-02-29	2025-01-13	2024-04-05
NCT03608033	Omeros Corporation	2024-01-12	2025-01-10	2024-03-12
NCT04432831	Hoffmann-La Roche	2023-10-11	2024-10-07	2024-08-23
NCT06437626	Pharmasoft	2023-08-18	2025-04-04	2024-07-12
NCT02807636	Hoffmann-La Roche	2022-08-31	2023-08-22	2023-06-08
NCT03083665	UCB Biopharma SRL	2022-06-30	2022-12-15	2022-10-06
NCT04478084	Sanofi Pasteur, a Sanofi Company	2022-03-19	(none)	2022-03-31
NCT05062330	Aldeyra Therapeutics, Inc.	2022-03-04	2025-02-27	2022-06-03
NCT03783442	BeiGene	2022-02-28	2023-10-10	2022-04-26
NCT02967692	Novartis Pharmaceuticals	2020-08-11	2023-06-26	2022-01-14
NCT02017717	Bristol-Myers Squibb	2019-06-17	2022-06-02	2020-01-09
NCT01358877	Hoffmann-La Roche	2016-12-19	2017-12-04	2017-04-20
NCT02677298	Croma-Pharma GmbH	2016-11-23	2024-12-17	2022-05-13
NCT01084876	Celltrion	2012-07	2024-10-29	2024-08-27
NCT00460512	Janssen-Cilag International NV	2009-01-22	(none)	2009-03-12
NCT00087776	American Regent, Inc.	2007-10-27	(none)	2025-07-21